

Time Series — Solutions

Practice Exam — Week 2: ARMA Models and the PACF

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Problem 1 (5 points)

Statement.

(a) State the definition of white noise and write down its ACVS γ_τ and ACF ρ_τ . Explain in one sentence why “uncorrelated” is weaker than “independent”. (1.5 pts)

(b) Consider the MA(2) process

$$X_t = \varepsilon_t - \theta_1 \varepsilon_{t-1} - \theta_2 \varepsilon_{t-2}.$$

Compute the mean and the autocovariance γ_τ for all $\tau \in \mathbb{Z}$, and write down ρ_1 and ρ_2 . (2.5 pts)

(c) For which lags τ does the ACVS of an MA(q) vanish? State the general rule and explain how it is used to read off the order q from an estimated ACF. (1 pt)

Solution.

(a) $\{\varepsilon_t\}$ is *white noise* if it is a sequence of mean-zero, finite-variance, *uncorrelated* random variables: $\mathbb{E}[\varepsilon_t] = 0$, $\text{Var}(\varepsilon_t) = \sigma^2 < \infty$, and $\text{Cov}(\varepsilon_s, \varepsilon_t) = 0$ for $s \neq t$. Its second-order structure is

$$\gamma_\tau = \begin{cases} \sigma^2, & \tau = 0, \\ 0, & \tau \neq 0, \end{cases} \quad \rho_\tau = \begin{cases} 1, & \tau = 0, \\ 0, & \tau \neq 0. \end{cases}$$

“Uncorrelated” only constrains the first two moments, whereas “independent” constrains the entire joint law; hence a white-noise process need not be i.i.d. (and need not be Gaussian).

(b) The constant mean is $\mathbb{E}[X_t] = \mathbb{E}[\varepsilon_t] - \theta_1 \mathbb{E}[\varepsilon_{t-1}] - \theta_2 \mathbb{E}[\varepsilon_{t-2}] = 0$. The filter taps are $(\psi_0, \psi_1, \psi_2) = (1, -\theta_1, -\theta_2)$. Using bilinearity and $\text{Cov}(\varepsilon_s, \varepsilon_u) = \sigma^2 \mathbf{1}_{\{s=u\}}$, the ACVS is $\gamma_\tau = \sigma^2 \sum_j \psi_j \psi_{j+|\tau|}$ (only matching noise indices survive):

$$\begin{aligned} \gamma_0 &= \sigma^2 (1 + \theta_1^2 + \theta_2^2), \\ \gamma_{\pm 1} &= \sigma^2 [(1)(-\theta_1) + (-\theta_1)(-\theta_2)] = \sigma^2 \theta_1 (\theta_2 - 1), \\ \gamma_{\pm 2} &= \sigma^2 [(1)(-\theta_2)] = -\sigma^2 \theta_2, \\ \gamma_\tau &= 0 \quad (|\tau| \geq 3). \end{aligned}$$

Dividing by γ_0 ,

$$\rho_1 = \frac{\theta_1(\theta_2 - 1)}{1 + \theta_1^2 + \theta_2^2}, \quad \rho_2 = \frac{-\theta_2}{1 + \theta_1^2 + \theta_2^2}.$$

(c) For an MA(q) with taps $\psi_0, \dots, \psi_q$ one has $\gamma_\tau = \sigma^2 \sum_j \psi_j \psi_{j+|\tau|}$, and for $|\tau| > q$ no pair of taps overlaps (the filter has length $q + 1$), so $\gamma_\tau = 0$ for $|\tau| > q$. Thus the **ACVS of an MA(q) cuts off after lag q** : the largest lag at which the (sample) ACF is still significant — beyond the $\pm 1.96/\sqrt{n}$ band — identifies the order q .

Problem 2 (5 points)

Statement.

Consider the AR(1) process $X_t = \phi X_{t-1} + \varepsilon_t$.

(a) State the condition on ϕ for the process to be causal, and explain it in terms of the root of the AR polynomial $\Phi(z) = 1 - \phi z$. (1 pt)

(b) Assuming $|\phi| < 1$, derive the causal MA(∞) representation $X_t = \sum_{k \geq 0} \psi_k \varepsilon_{t-k}$ by iterating the recursion. Give the ψ -weights and verify $\sum_k |\psi_k| < \infty$. (2 pts)

(c) Using the representation in (b), show that

$$\gamma_\tau = \frac{\sigma^2}{1 - \phi^2} \phi^{|\tau|}, \quad \rho_\tau = \phi^{|\tau|}.$$

Why does this ACF tail off rather than cut off, and what does that say about identifying p from the ACF? (2 pts)

Solution.

(a) The process is *causal* iff $|\phi| < 1$. The AR polynomial $\Phi(z) = 1 - \phi z$ has its single root at $z = 1/\phi$, and the causality condition “root outside the unit circle” reads $|1/\phi| > 1 \iff |\phi| < 1$.

(b) Iterating $X_t = \phi X_{t-1} + \varepsilon_t$, with $X_{t-1} = \phi X_{t-2} + \varepsilon_{t-1}$, etc.:

$$\begin{aligned} X_t &= \varepsilon_t + \phi X_{t-1} = \varepsilon_t + \phi \varepsilon_{t-1} + \phi^2 X_{t-2} \\ &= \varepsilon_t + \phi \varepsilon_{t-1} + \phi^2 \varepsilon_{t-2} + \cdots + \phi^{n-1} \varepsilon_{t-n+1} + \phi^n X_{t-n}. \end{aligned}$$

Since $|\phi| < 1$ and $\text{Var}(X_{t-n}) = \gamma_0$ is constant, the remainder satisfies $\mathbb{E}[(\phi^n X_{t-n})^2] = \phi^{2n} \gamma_0 \rightarrow 0$, so it vanishes in mean square and

$$X_t = \sum_{k=0}^{\infty} \phi^k \varepsilon_{t-k}, \quad \psi_k = \phi^k.$$

The weights are absolutely summable: $\sum_{k \geq 0} |\psi_k| = \sum_{k \geq 0} |\phi|^k = \frac{1}{1 - |\phi|} < \infty$, confirming a valid causal MA(∞) representation.

(c) With $\mathbb{E}[X_t] = 0$ and $\text{Cov}(\varepsilon_{t-k}, \varepsilon_{t-\tau-l}) = \sigma^2 \mathbf{1}_{\{t-k=t-\tau-l\}}$, take $\tau \geq 0$:

$$\begin{aligned} \gamma_\tau &= \text{Cov}\left(\sum_{k \geq 0} \phi^k \varepsilon_{t-k}, \sum_{l \geq 0} \phi^l \varepsilon_{t-\tau-l}\right) = \sigma^2 \sum_{l \geq 0} \phi^{l+\tau} \phi^l \\ &= \sigma^2 \phi^\tau \sum_{l \geq 0} \phi^{2l} = \frac{\sigma^2 \phi^\tau}{1 - \phi^2}, \end{aligned}$$

where the surviving terms are those with $k = l + \tau$, and the geometric series converges since $\phi^2 < 1$. By symmetry $\gamma_\tau = \frac{\sigma^2}{1 - \phi^2} \phi^{|\tau|}$, and $\rho_\tau = \gamma_\tau / \gamma_0 = \phi^{|\tau|}$.

The ACF $\phi^{|\tau|}$ decays geometrically but is never exactly 0 for $|\phi| \in (0, 1)$, so it *tails off* rather than cuts off. Consequently the order p of an AR cannot be read from the ACF; it is the **PACF** that cuts off at lag p (here $p = 1$).

Problem 3 (5 points)

Statement.

(a) Consider the AR(2) process $X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \varepsilon_t$ with $\phi_1 = \frac{1}{2}$, $\phi_2 = -\frac{1}{4}$. Using the causality conditions $\phi_1 + \phi_2 < 1$, $\phi_2 - \phi_1 < 1$, $|\phi_2| < 1$, verify that the process is causal. (1 pt)

(b) From the Yule–Walker equations $\rho_\tau = \phi_1 \rho_{\tau-1} + \phi_2 \rho_{\tau-2}$ (with $\rho_0 = 1$, $\rho_{-1} = \rho_1$), compute ρ_1, ρ_2, ρ_3 . (2 pts)

(c) Using the Durbin–Levinson recursion

$$\alpha_{1,1} = \rho_1, \quad \alpha_{\tau,\tau} = \frac{\rho_\tau - \sum_{j=1}^{\tau-1} \alpha_{\tau-1,j} \rho_{\tau-j}}{1 - \sum_{j=1}^{\tau-1} \alpha_{\tau-1,j} \rho_j}, \quad \alpha_{\tau,j} = \alpha_{\tau-1,j} - \alpha_{\tau,\tau} \alpha_{\tau-1,\tau-j},$$

compute $\alpha_1, \alpha_2, \alpha_3$ from the values in (b). Read off the implied model order and state the ACF/PACF identification rule it illustrates. (2 pts)

Solution.

(a) With $\phi_1 = \frac{1}{2}$, $\phi_2 = -\frac{1}{4}$:

$$\phi_1 + \phi_2 = \frac{1}{4} < 1, \quad \phi_2 - \phi_1 = -\frac{3}{4} < 1, \quad |\phi_2| = \frac{1}{4} < 1.$$

All three causality conditions hold, so the AR(2) is causal (both roots of $\Phi(z) = 1 - \frac{1}{2}z + \frac{1}{4}z^2$ lie outside the unit circle).

(b) The Yule–Walker recursion $\rho_\tau = \phi_1\rho_{\tau-1} + \phi_2\rho_{\tau-2}$. At $\tau = 1$, using $\rho_{-1} = \rho_1$, we get $\rho_1 = \phi_1\rho_0 + \phi_2\rho_1 = \phi_1 + \phi_2\rho_1$, so

$$\rho_1 = \frac{\phi_1}{1 - \phi_2} = \frac{1/2}{1 + 1/4} = \frac{1/2}{5/4} = \frac{2}{5}.$$

Then

$$\begin{aligned} \rho_2 &= \phi_1\rho_1 + \phi_2 = \frac{1}{2} \cdot \frac{2}{5} - \frac{1}{4} = \frac{1}{5} - \frac{1}{4} = -\frac{1}{20}, \\ \rho_3 &= \phi_1\rho_2 + \phi_2\rho_1 = \frac{1}{2} \cdot \left(-\frac{1}{20}\right) - \frac{1}{4} \cdot \frac{2}{5} = -\frac{1}{40} - \frac{1}{10} = -\frac{1}{40} - \frac{4}{40} = -\frac{5}{40} = -\frac{1}{8}. \end{aligned}$$

(c) Durbin–Levinson with $\rho_1 = \frac{2}{5}$, $\rho_2 = -\frac{1}{20}$, $\rho_3 = -\frac{1}{8}$.

Lag 1: $\alpha_{1,1} = \rho_1 = \frac{2}{5}$.

Lag 2:

$$\alpha_{2,2} = \frac{\rho_2 - \alpha_{1,1}\rho_1}{1 - \alpha_{1,1}\rho_1} = \frac{-\frac{1}{20} - \frac{2}{5} \cdot \frac{2}{5}}{1 - \frac{2}{5} \cdot \frac{2}{5}} = \frac{-\frac{1}{20} - \frac{4}{25}}{1 - \frac{4}{25}} = \frac{-\frac{21}{100}}{\frac{84}{100}} = -\frac{21}{84} = -\frac{1}{4}.$$

Update: $\alpha_{2,1} = \alpha_{1,1} - \alpha_{2,2}\alpha_{1,1} = \frac{2}{5} - \left(-\frac{1}{4}\right)\frac{2}{5} = \frac{2}{5} + \frac{1}{10} = \frac{1}{2}$.

Lag 3:

$$\alpha_{3,3} = \frac{\rho_3 - \alpha_{2,1}\rho_2 - \alpha_{2,2}\rho_1}{1 - \alpha_{2,1}\rho_1 - \alpha_{2,2}\rho_2}.$$

The numerator is

$$-\frac{1}{8} - \frac{1}{2} \cdot \left(-\frac{1}{20}\right) - \left(-\frac{1}{4}\right) \cdot \frac{2}{5} = -\frac{1}{8} + \frac{1}{40} + \frac{1}{10} = -\frac{5}{40} + \frac{1}{40} + \frac{4}{40} = 0,$$

so $\alpha_{3,3} = 0$ (the denominator $1 - \frac{1}{2} \cdot \frac{2}{5} - \left(-\frac{1}{4}\right)\left(-\frac{1}{20}\right) = 1 - \frac{1}{5} - \frac{1}{80} > 0$ is nonzero). Hence

$$\alpha_1 = \frac{2}{5}, \quad \alpha_2 = -\frac{1}{4}, \quad \alpha_3 = 0.$$

The PACF cuts off after lag 2 while the ACF in (b) tails off: this is the **AR(p) signature with $p = 2$** . Indeed the diagonal coefficients recover the AR coefficients, $\alpha_{2,1} = \frac{1}{2} = \phi_1$ and $\alpha_{2,2} = -\frac{1}{4} = \phi_2$. (Identification rule: PACF cuts after lag p , ACF tails off $\Rightarrow$ AR(p).)

Problem 4 (5 points)

Statement.

(Revision of Week 1: stationarity, the ACVS & ergodicity.) Let $\{Y_t\}$ be i.i.d. with mean 0 and variance σ_Y^2 , and define $X_t = Y_t + \frac{1}{2}Y_{t-1}$.

(a) Compute $\mathbb{E}[X_t]$ and the autocovariance γ_τ^X for all $\tau \in \mathbb{Z}$, give the lag-1 autocorrelation ρ_1 , and explain in one sentence why $\{X_t\}$ is weakly stationary. (2 pts)

(b) (Validity via positive semi-definiteness.) A candidate sequence is a valid ACVS of some stationary process only if it is positive semi-definite. Decide whether $\gamma_0 = 1$, $\gamma_{\pm 1} = 0.8$, $\gamma_\tau = 0$ ($|\tau| \geq 2$) is a valid ACVS, by forming the 3×3 covariance matrix $\Gamma_3 = [\gamma_{j-k}]_{1 \leq j, k \leq 3}$ and testing its definiteness. (2 pts)

(c) (Mean ergodicity.) Show that if $\sum_{\tau} |\gamma_{\tau}| < \infty$ then $\text{Var}(\bar{X}_n) \rightarrow 0$, where $\bar{X}_n = \frac{1}{n} \sum_{t=1}^n X_t$, so $\bar{X}_n$ is a consistent (mean-ergodic) estimator of $\mu = \mathbb{E}[X_t]$. (1 pt)

Solution.

(a) Since $\mathbb{E}[Y_t] = 0$, the mean is constant: $\mathbb{E}[X_t] = \mathbb{E}[Y_t] + \frac{1}{2}\mathbb{E}[Y_{t-1}] = 0$. The Y 's are uncorrelated with variance σ_Y^2 , so

$$\gamma_0^X = \text{Var}(X_t) = \text{Var}(Y_t) + \frac{1}{4} \text{Var}(Y_{t-1}) = \left(1 + \frac{1}{4}\right) \sigma_Y^2 = \frac{5}{4} \sigma_Y^2,$$

$$\gamma_1^X = \text{Cov}(Y_t + \frac{1}{2}Y_{t-1}, Y_{t-1} + \frac{1}{2}Y_{t-2}) = \frac{1}{2} \text{Var}(Y_{t-1}) = \frac{1}{2} \sigma_Y^2,$$

and $\gamma_{\tau}^X = 0$ for $|\tau| \geq 2$ (no shared Y index). Hence

$$\rho_1 = \frac{\gamma_1^X}{\gamma_0^X} = \frac{1/2}{5/4} = \frac{2}{5} = 0.4.$$

The mean is constant and γ_{τ}^X depends only on the lag τ (not on t), with $\gamma_0^X < \infty$; therefore $\{X_t\}$ is weakly stationary.

(b) The covariance matrix of (X_1, X_2, X_3) is the symmetric Toeplitz matrix

$$\Gamma_3 = \begin{pmatrix} 1 & 0.8 & 0 \\ 0.8 & 1 & 0.8 \\ 0 & 0.8 & 1 \end{pmatrix}.$$

Positive semi-definiteness requires *every* such covariance matrix Γ_n to have a non-negative determinant. Expanding along the first row,

$$\det \Gamma_3 = 1 \cdot (1 - 0.8^2) - 0.8 \cdot (0.8 - 0) + 0 = (1 - 0.64) - 0.64 = -0.28 < 0.$$

A positive-semi-definite matrix cannot have a negative determinant, so Γ_3 is *not* positive semi-definite (it has a negative eigenvalue). Hence the sequence is **not** a valid ACVS of any stationary process. (Intuitively a lag-1 correlation of 0.8 with zero correlation beyond lag 1 is “too strong”: contrast part (a), whose genuine model only reaches $\rho_1 = 0.4$.)

(c) Expand the variance of the sample mean and group the $n - |\tau|$ pairs at each lag τ :

$$\text{Var}(\bar{X}_n) = \frac{1}{n^2} \sum_{s=1}^n \sum_{t=1}^n \text{Cov}(X_s, X_t) = \frac{1}{n^2} \sum_{s,t} \gamma_{s-t} = \frac{1}{n} \sum_{\tau=-(n-1)}^{n-1} \left(1 - \frac{|\tau|}{n}\right) \gamma_{\tau}.$$

Bounding the weights by $0 \leq 1 - |\tau|/n \leq 1$,

$$\text{Var}(\bar{X}_n) \leq \frac{1}{n} \sum_{\tau=-(n-1)}^{n-1} |\gamma_{\tau}| \leq \frac{1}{n} \sum_{\tau \in \mathbb{Z}} |\gamma_{\tau}| = \frac{C}{n} \xrightarrow{n \rightarrow \infty} 0,$$

with $C = \sum_{\tau} |\gamma_{\tau}| < \infty$ by hypothesis. Since $\mathbb{E}[\bar{X}_n] = \mu$, this means $\bar{X}_n \rightarrow \mu$ in mean square, i.e. $\bar{X}_n$ is a consistent (mean-ergodic) estimator of μ .